

INTERSTITIAL NEPHRITIS IN THE SPINY MOUSE (*ACOMYS CAHIRINUS*) ASSOCIATED WITH *KLOSSIELLA* SP. INFECTION

by

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SUMMARY

Klossiella sp. infection in the spiny mouse (*Acomys cahirinus*) was found associated with interstitial nephritis. The clinical disease appeared to be related to unsanitary housing conditions, and disappeared when hygiene was improved.

INTERSTITIELLE NEPHRITIS IN DER STACHELIGEN MAUS (*ACOMYS CAHIRINUS*) MIT *KLOSSIELLA* SP.-INFEKTION VERBUNDEN

Man fand eine Beziehung zwischen *Klossiella* sp. Infektion in der stacheligen Maus (*Acomys cahirinus*) und interstitieller Nephritis. Die klinische Krankheit schien mit unhygienischen Behausungsbedingungen zusammenzuhängen und verschwand, wenn die Hygiene verbessert wurde.

NEPHRITE INTERSTITIELLE DANS LA SOURIS AUX EPINES (*ACOMYS CAHIRINUS*) ASSOCIEE A L'INFECTION AVEC *KLOSSIELLA* SP.

On a trouvé une relation entre l'infection avec *Klossiella* sp. dans la souris épinée (*Acomys cahirinus*) et la néphrite interstitielle. La maladie clinique semblait être associée à des conditions de domicile non-hygiéniques, et disparaissait lorsque l'hygiène fut améliorée.

A few spiny mice (*Acomys cahirinus*) which were captured in the Judean desert (Israel) were bred in the laboratory for several months. Sickness was then observed in the colony; the animals were apathetic and underweight, and sporadic death occurred. The colony consisted of 350 mice of both sexes and various ages: 52 of these died within 4 weeks. The disease developed during a period when animal husbandry was poor, and the clinical disease disappeared shortly after the animals were transferred to clean, dry cages.

7 animals of the infected colony were examined when in a moribund state, or immediately after death. Laboratory examinations were performed, and *Klossiella* sp. infection was found in the kidneys of all of these animals.

Klossiella is a protozoan classified as follows (Levine & Ivens, 1965):

Subphylum: Sporozoa. Class: Telosporasida.

Subclass: Coccidiasina. Order: Eucoccidiorida.

Suborder: Adeleidae. Family: Klossiellidae.

LABORATORY EXAMINATIONS

Bacteriology

Samples of heart, liver, spleen, small and large intestines were cultured aerobically on blood agar, eosin methylene blue agar, and tryptic soy agar, and anaerobically on dextrose-blood-agar, dextrose-plasma-agar, tomato juice agar and fluid thioglycollate medium, at 37°C for 24–48 hours. No growth was observed.

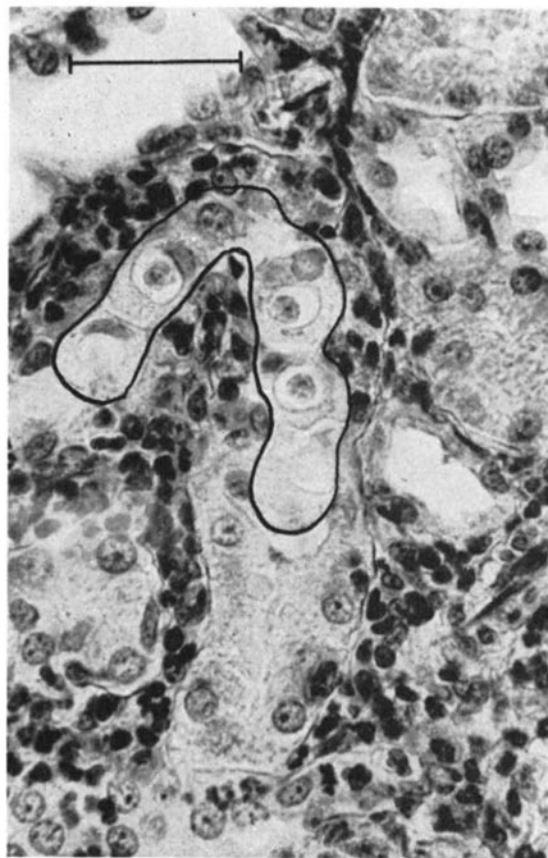


Fig. 1. Kidney cortex. Macrogametes can be seen in 4 epithelial cells of a convoluted tubule. Interstitial inflammation is present. Line represents 50 μ m.

Parasitology

Ectoparasites. On all animals examined, a few mites (*Myocoptes musculus*) were found.

Endoparasites. In the caecum of all animals examined, nematodes (*Syphacia obvelata*) and protozoa (*Trichomonas* sp.) were found. No blood parasites were found.

Histopathology

Histological sections were prepared of heart, lungs, intestines, liver, spleen, kidneys and adrenals. Lungs, intestines, liver, spleen and adrenals showed no pathological changes. In 2 animals *Sarcocystis* was observed in the heart muscle without any associated tissue reaction. All the examined kidneys showed similar lesions: diffuse and focal infiltration by mononuclear cells, mainly around the tubules and slightly around the glomeruli; these chronic inflammatory areas also contained debris. In the lumen of the tubules many granulocytes were seen. A few parasites were situated within the tubules without evidence of associated reaction, but many parasites were seen within the areas of inflammation and debris (see Figs 1 and 4).

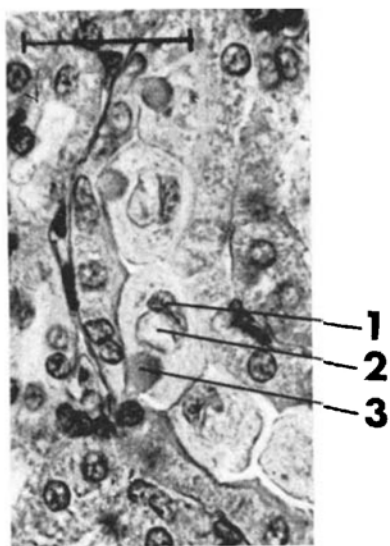


Fig. 2. Kidney cortex. Macrogametes in renal tubule cells. 1. nucleus of the epithelial cell. 2. macrogamete. 3. region of penetration of the parasite.

Line represents 50 μ m.

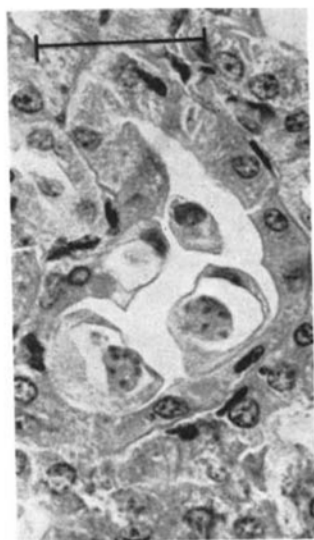


Fig. 3. 2 sporonts in a renal tubule (above which a macrogamete in an epithelial cell can be seen).

Line represents 50 μ m.

In our opinion these parasites are *Klossiella* sp. This diagnosis was made by comparing our material with the description given by Cohrs, Jaffe & Meessen (1958) and with specimens in the collection of the U.S. Armed Forces Institute of Pathology (Slide # 15, 60-10F1; comments by R. M. McCully, Capt. VL, USAF).

Histological material was also sent to Professor N. D. Levine of the University of Illinois, U.S.A., who confirmed this diagnosis.

According to Levine & Ivens (1965), infection takes place by ingestion of sporocysts which reach the kidneys by way of the blood stream as sporozoites. There are 2 types of schizogony, of which the more common occurs in the arterioles and capillaries of the kidneys and in the epithelium of the Bowman's capsule; in this type 40-60 merozoites are formed. The mature schizont breaks, releasing the merozoites into the lumen of the kidney tubule, from which they enter the epithelial cells of the convoluted tubules. Here gamonts

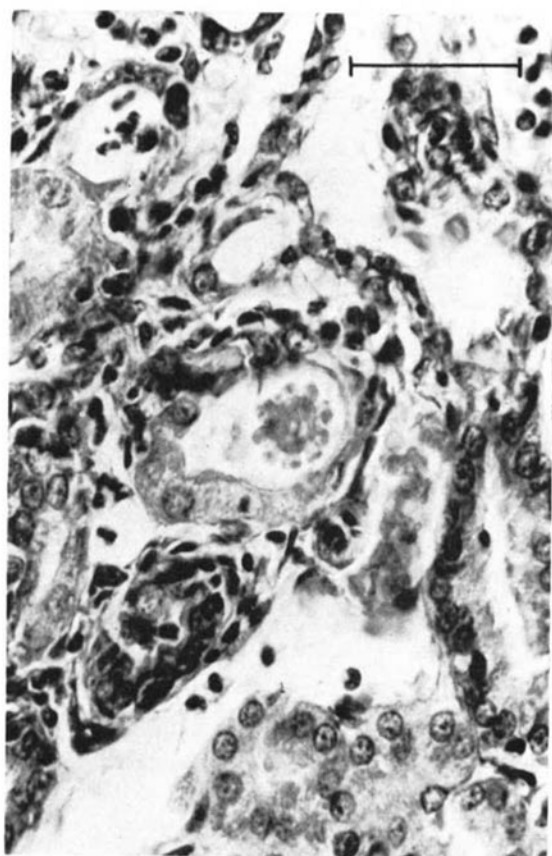


Fig. 4. A sporocyst with sporozoites in a renal tubule. Note the intestinal inflammation. Line represents 50 μ m.

are formed and gametogony and sporogony take place. Macrogametes (Figs 1 and 2) and microgametocytes occur together within a vacuole in the host cell; the microgametocyte divides to form 2 microgametes, and fertilization takes place to form a zygote or sporont. The sporont (2 of them can be seen in Fig. 3) contains plastin granules, and divides to form sporoblasts. The resulting sporocyst is spherical (Fig. 4). The sporozoites lie side by side in the sporocyst.

The sporocysts are released into the lumen of the kidney tubules by rupture of the host cell (Fig. 5) and pass out in the urine.

Unfortunately, infection of healthy spiny mice with urine of diseased animals could not be attempted because of lack of adequate material.

The animals were transferred into new, clean, suitable cages; appropriate mouse diet was given and an optimum temperature control was initiated. A week later the improved husbandry had resulted in the disappearance of clinical signs.

A group of infected spiny mice was again subjected to conditions of high humidity, inadequate food and uncontrolled temperature, and 7 days later the same clinical signs began to appear. Upon examination the same laboratory findings were observed.



Fig. 5. Sporozoites within the lumen of a renal tubule. Line represents 50 μ m.

DISCUSSION

Klossiella is a protozoan parasite which is common in *Mus musculus* and is normally non-pathogenic in the kidney. The disease occurred during a

period when animal husbandry was very poor: cage bedding was not changed often enough, the animals were wet and dirty, temperature varied between 10 and 30°C, and the food was inadequate. The clinical disease disappeared shortly after the animals were transferred to clean, dry cages. Although *Klossiella* was found in kidneys of mice examined 2 weeks later, no inflammatory changes were then observed. It seems that a latent infection by *Klossiella* can result in nephritis under conditions of poor husbandry. This type of infection has not previously been described in spiny mice.

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